作业纸

班级：　　姓名：　　学号：　　科目：　　第　页

4.27 虹吸利用大气压强将水压入虹吸管

对质量为Δm的水分析

伯努利方程：　$\rho g h = \frac{1}{2}\rho v^2$

流量 $Q = Sv = \pi r^2 v$

解 $Q = 12.43\ m^3/s = 4.47\times10^3\ m^3/h$

即每小时引入田地的河水体积为 $4.47\times10^3\ m^3$

4.28 伯努利方程

$P_0 = P_1 + \frac{1}{2}\rho v^2$

$P_0 = 21\ atm$

$P_1 = 1\ atm$　　$Q = \frac{\pi}{4}d^2 v$

解：$Q = 3198\ cm^3/s = 1.92\times10^5\ cm^3/min$

即每分钟喷出杀虫剂水的体积为 $1.92\times10^5\ cm^3$

5.3 1900年地球自转角速度分别为ω_1，ω_2

1987

由题知 $365\left(\frac{2\pi}{\omega_2}-\frac{2\pi}{\omega_1}\right) = \Delta t$

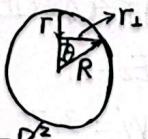
$\bar{\alpha} = \frac{\omega_2-\omega_1}{87\Delta T}$

$\Delta T \approx 365\frac{2\pi}{\omega_1} \approx 3.15\times10^7\ s$

解 $\bar{\alpha} = -\frac{365\times2\times3.14\times1.14}{87\times(3.15\times10^7)^3}\ rad/s^2$

$= -9.6\times10^{-22}\ rad/s^2$

5.6 (1) $J = \sum m r_\perp^2$



~~由几何关系有 $(R-r)^2+r_\perp^2=R^2$~~

~~即 $r^2-2Rr+r_\perp^2=0$~~

设$\vec{r}$与$\vec{R}$的夹角为$(\pi-\theta)$ 则 $r_\perp = R\sin\theta$

$r = R(1-\cos\theta)$

$J = \int_0^\pi \rho\cdot2\pi R^2\sin\theta\, d\theta\cdot(R\sin\theta)^2$

$= 2\pi\rho R^4\int_0^\pi \sin^3\theta\, d\theta$

$= \frac{8}{3}\pi\rho R^4$

又 $\rho\cdot4\pi R^2 = m$

解 $J = \frac{2}{3}mR^2$

其中 $m = 60\times12\times1.66\times10^{-27}kg$

$R = \frac{71}{2}\times10^{-9}m$

解 $J = 1.004\times10^{-39}\ kg\cdot m^2$

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(2) $\frac{1}{2}J\omega^2 = 6.21\times10^{-21}\,J$

$f = \frac{\omega}{2\pi}$

解 $f = 5.60\times10^{8}\,Hz$

5.10. 设薄板的面密度为ρ

$\rho\left[\pi R^2 - \pi\left(\frac{R}{2}\right)^2\right] = m$

所剩薄板质心距O点为L则

$mL = \rho\pi\left(\frac{R}{2}\right)^2\cdot\frac{R}{2}$

解 $L = \frac{R}{6}$

完整薄板的转动惯量

$J = \frac{1}{2}\left(m+\frac{m}{3}\right)R^2$

以O'为旋转轴，被挖去的圆板的转动惯量

$J_2 = \frac{1}{2}\cdot\frac{m}{3}R^2$

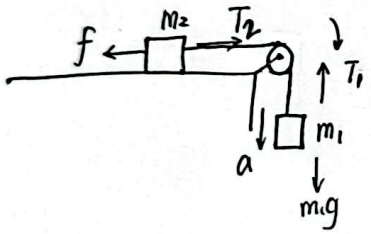
由平行轴定理有

$J_2' = J_2 + \frac{m}{3}\left(\frac{R}{2}\right)^2$

转动惯量可叠加 则 $J = J_1 + J_2'$

解 $J_1 = \frac{13}{24}mR^2$

5.11



对 m_1 受力分析有

$m_1g - T_1 = m_1a$

对 m_2 受力分析有

$T_2 - f = m_2a$

$f = \mu_k m_2 g$

对滑轮分析

$T_2 r + J\alpha = T_1 r$

$\alpha = \frac{a}{r}$

$J = \frac{1}{2}mr^2$

解：
$$\begin{cases} a = \dfrac{m_1g - \mu_k m_2 g}{m_1 + m_2 + \frac{m}{2}} \\ T_1 = m_1 g\,\dfrac{(1+\mu_k)m_2 + \frac{m}{2}}{m_1 + m_2 + \frac{m}{2}} \\ T_2 = m_2 g\,\dfrac{(1+\mu_k)m_1 + \frac{\mu_k m}{2}}{m_1 + m_2 + \frac{m}{2}} \end{cases}$$

5.15. $n_1 = 15\,r/min = \frac{1}{4}\,r/s = \frac{\omega_1}{2\pi}$

同理 $n_2 = \frac{\omega_2}{2\pi}$

由角动量守恒可得 $J_1\omega_1 = J_2\omega_2$

解 $n_2 = 37.5\,r/min$

$E_1 = \frac{1}{2}J_1\omega_1^2$

$E_2 = \frac{1}{2}J_2\omega_2^2$

$\frac{E_1}{E_2} = \frac{\omega_1}{\omega_2} = \frac{2}{5} \neq 1$

可知系统的机械能不守恒

人肌肉的张力做正功

$W = E_2 - E_1$

解 $W = 3.70\,J$

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